

CYBER SECURITY TASK 01

PART B – PASSWORD SECURITY REVIEW

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Introduction :

Passwords are the first line of defense for protecting online accounts and personal information. Weak or reused passwords can make it easier for attackers to gain unauthorized access to sensitive data. By following strong password practices and enabling additional security measures such as Multi-Factor Authentication (MFA), users can significantly reduce the risk of cyberattacks. This section explains the importance of password security and outlines best practices for creating and managing secure passwords.

1. What makes a strong password?

A strong password is one that is difficult to guess or crack. It should be at least 12 characters long and include a combination of uppercase letters, lowercase letters, numbers, and special characters. A strong password should not contain personal information such as names, birthdays, or phone numbers. Using random and unique combinations of characters makes the password more secure against brute-force and dictionary attacks.

2. Why should passwords be unique?

Each online account should have a unique password to prevent multiple accounts from being compromised if one password is exposed. If the same password is used across different websites and one website experiences a data breach, attackers can use the stolen credentials to access other accounts. Unique passwords help limit the impact of security incidents and improve overall account protection.

3. What is Multi-Factor Authentication (MFA)?

Multi-Factor Authentication (MFA) is a security method that requires users to verify their identity using two or more authentication factors before accessing an

account. These factors may include a password, a one-time verification code (OTP), a fingerprint, facial recognition, or an authentication app. MFA adds an extra layer of security, making it much harder for attackers to gain unauthorized access even if they know the password.

4. What are the risks of password reuse?

Reusing the same password across multiple accounts increases the risk of unauthorized access. If one account is compromised due to a data breach, attackers may attempt to use the same credentials on other websites, a technique known as credential stuffing. This can result in multiple accounts being hacked, leading to identity theft, financial loss, and exposure of personal information.

Password Policy (Best Practices) :

1. Use passwords that are at least 12 characters long.
2. Include a combination of uppercase letters, lowercase letters, numbers, and special characters.
3. Avoid using personal information such as names, birthdays, or phone numbers.
4. Use a unique password for every online account.
5. Enable Multi-Factor Authentication (MFA) whenever it is available.
6. Store passwords securely using a trusted password manager.
7. Change passwords immediately if you suspect they have been compromised.
8. Do not share passwords with anyone or store them in unsecured locations.

Conclusion :

Strong password practices are essential for protecting online accounts and sensitive information from cyber threats. Using unique and complex passwords, together with Multi-Factor Authentication, greatly improves account security and reduces the risk of unauthorized access. Following a well-defined password policy helps maintain good cyber hygiene and strengthens overall digital security.